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$T_p\mathbb{R}^n$ es isomorfo a $\mathbb{R}^n$

Teorema 1 ($T_p\mathbb{R}^n \cong \mathbb{R}^n$). *$\forall p \in \mathbb{R}^n : \phi \colon \mathbb{R}^n \longrightarrow T_p(\mathbb{R}^n)$ es un isomorfismo.*
$$v \longmapsto D_v|_p$$

***Demostración*:** Basta probar que ϕ es lineal y biyectiva:

- Linealidad: Sean $a, b \in \mathbb{R}$ y $u, v \in \mathbb{R}^n$, entonces $\forall f \in \mathcal{C}^\infty(\mathbb{R}^n)$:
$$\begin{aligned}\phi(au+bv)(f) = D_{au+bv}|_p(f) &= \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} f(p+t(au+bv)) \\ &= a \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} f(p+tu) + b \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} f(p+tv) = a\, D_u|_p(f) + b\, D_v|_p(f) \\ &= a\phi(u)(f) + b\phi(v)(f) \implies \phi(au+bv) = a\phi(u) + b\phi(v).\end{aligned}$$

- Inyectividad: Sean $u, v \in \mathbb{R}^n$ tales que $\phi(u) = \phi(v)$. Entonces $\forall f \in \mathcal{C}^\infty(\mathbb{R}^n)$ tenemos
$$D_u|_p(f) = \sum_{i=1}^n u_i \left.\frac{\partial}{\partial x_i}\right|_p (f) = \sum_{i=1}^n v_i \left.\frac{\partial}{\partial x_i}\right|_p (f) = D_v|_p(f).$$
Por tanto, al tomar $f = x^j$ (la proyección sobre la j-ésima coordenada), tenemos que
$$\forall j \in \mathbb{N}_n : D_u|_p(x^j) = \sum_{i=1}^n u_i \left.\frac{\partial}{\partial x_i}\right|_p (x^j) = u_j \underbrace{\left.\frac{\partial}{\partial x_j}\right|_p (x^j)}_{1} = u_j = v_j \implies u = v.$$

- Sobreyectividad: Sea $w \in T_p(\mathbb{R}^n)$, buscamos $v = (v_1, \ldots, v_n) \in \mathbb{R}^n$ tal que $\phi(v) = w$. Definimos $\forall i \in \mathbb{N}_n : v_i = w(x^i)$, donde x^i es la proyección sobre la i-ésima coordenada.

  Sea $f \in \mathcal{C}^\infty(\mathbb{R}^n)$, entonces, por el teorema de Taylor, tenemos que
$$f(x) = f(p) + \sum_{i=1}^n \left.\frac{\partial}{\partial x^i}\right|_p (f) \cdot (x_i - p_i) + \varphi(x) \sum_{i,j=1}^n (x_i - p_i)(x_j - p_j).$$
  Como $w\left(\varphi \cdot \sum_{i,j=1}^n (x^i - p_i)(x^j - p_j)\right) = \sum_{i,j=1}^n w\Big(\underbrace{\varphi \cdot (x^i - p_i)(x^j - p_j)}_{\text{se anula en } p}\Big)$, por (2) del

`lem-derivacion/`?? se tiene que $w\left(\varphi \cdot \sum_{i,j=1}^{n} (x^i - p_i)(x^j - p_j)\right) = 0$.

$$\implies w(f) = \cancel{w(f(p))} + \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(p) \underbrace{(w(x^i) - w(p_i))}_{v_i} + \cancel{w\left(\varphi \cdot \sum_{i,j=1}^{n} (x^i - p_i)(x^j - p_j)\right)}$$

$$= \sum_{i=1}^{n} v_i \cdot \frac{\partial f}{\partial x_i}(p) = \left. D_v \right|_p (f) \implies w = \left. D_v \right|_p = \phi(v).$$

Como ϕ es lineal, inyectiva y sobreyectiva, concluimos que es un isomorfismo. ■

Referenciado en

- `cor-dim-esp-tangente-variedad`

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